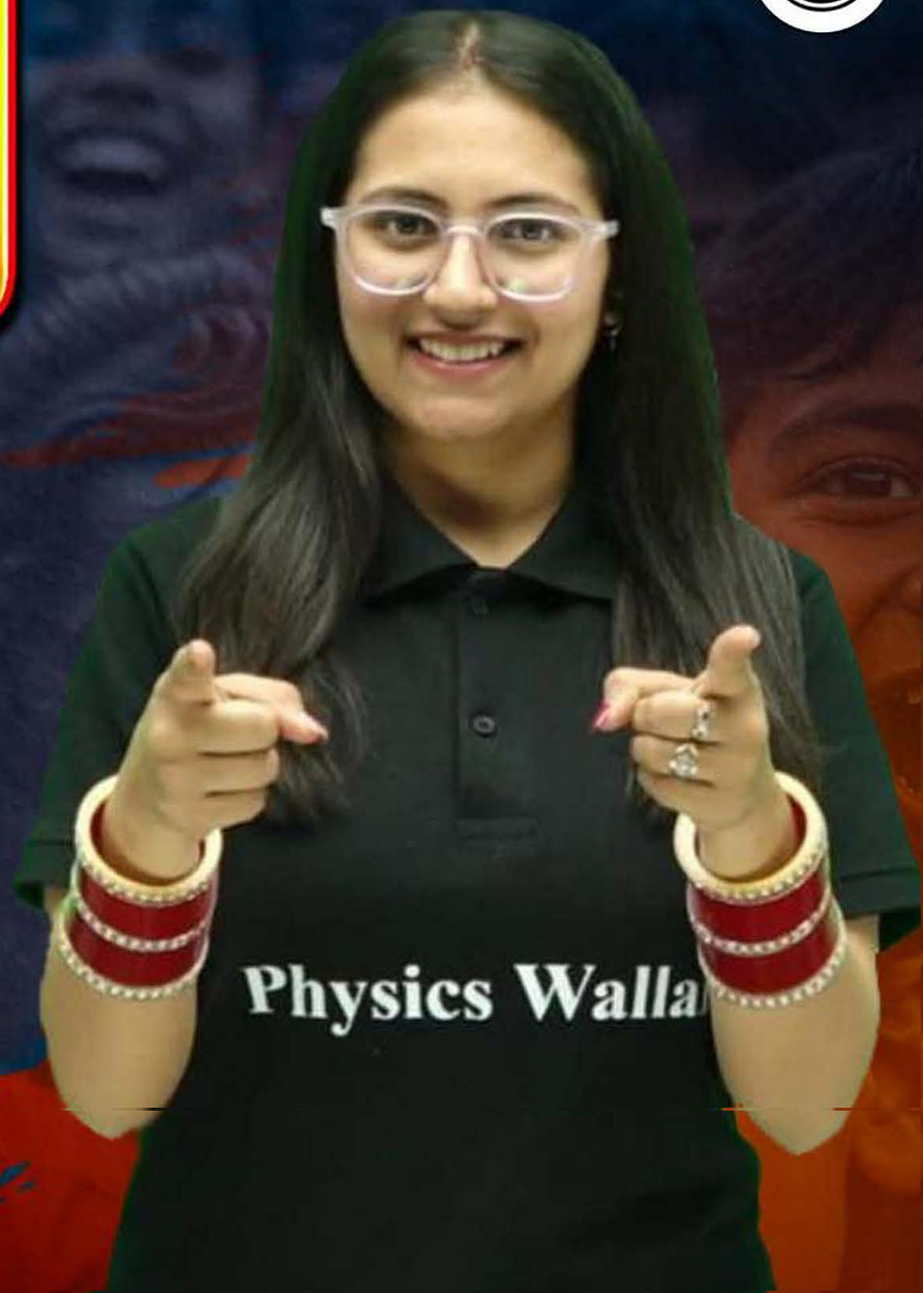
VICTORY 2026

CHEMISTRY

Study of Compounds

Lect.-01

By- Bharti Ma'am



Topics *to be covered*

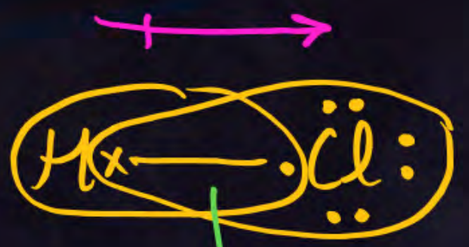
Study of compound: HCl

1. Prepⁿ of HCl

   Properties

HCl

$F > O > N \simeq Cl$



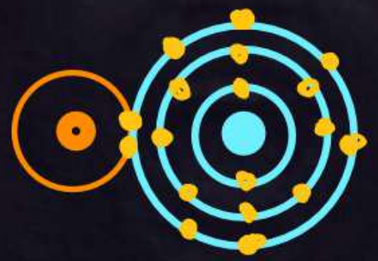
Sharing of one pair of e^-

* Type of Bond: Covalent Bond
* Polar Covalent Compound
* Molecular Mass = 1u + 35.5u
  = 36.5u

* $NaCl + H_2SO_4$ (Acid) → HCl

* Sir Lavoisier, Named Muriatic acid.
- HCl: Hydrochloric acid

Orbital diagram of HCl



H = K(1) Cl = 2,8,7

* HCl is present in our body, to provide acidic Medium for the activation of gastric enzymes

* In free form HCl is present in volcanic erruptions.

General preparation of hydrogen chloride gas

1. By synthesis (Direct combination)

→ H_2 gas directly reacts with Cl_2 gas in the presenceaf diffused sunlight

$$H_2(g) + Cl_2(g) \xrightarrow[\text{Sunlight}]{\text{Diffused}} 2HCl(g)$$ (Direct Combination reaction)

#B.M Vision:→ Above reaction become explosive in direct sunlight

General preparation of hydrogen chloride gas

2. By heating metallic chloride with conc. Sulphuric acid.

Metal Chloride + Sulphuric $\rightarrow$

$$NaCl + H_2SO_4 \xrightarrow{<200^\circ C} NaHSO_4 + HCl$$

$$CuCl_2 + H_2SO_4 \xrightarrow{\Delta} CuSO_4 + HCl$$

$$NaCl + H_2SO_4 \xrightarrow[200^\circ C]{\text{Below}} NaHSO_4 + HCl$$

$$2\,NaCl + H_2SO_4 \xrightarrow[200^\circ C]{\text{Above}} Na_2SO_4 + 2HCl$$

Imp

Laboratory preparation of hydrogen chloride gas

Reactants:

Imp.

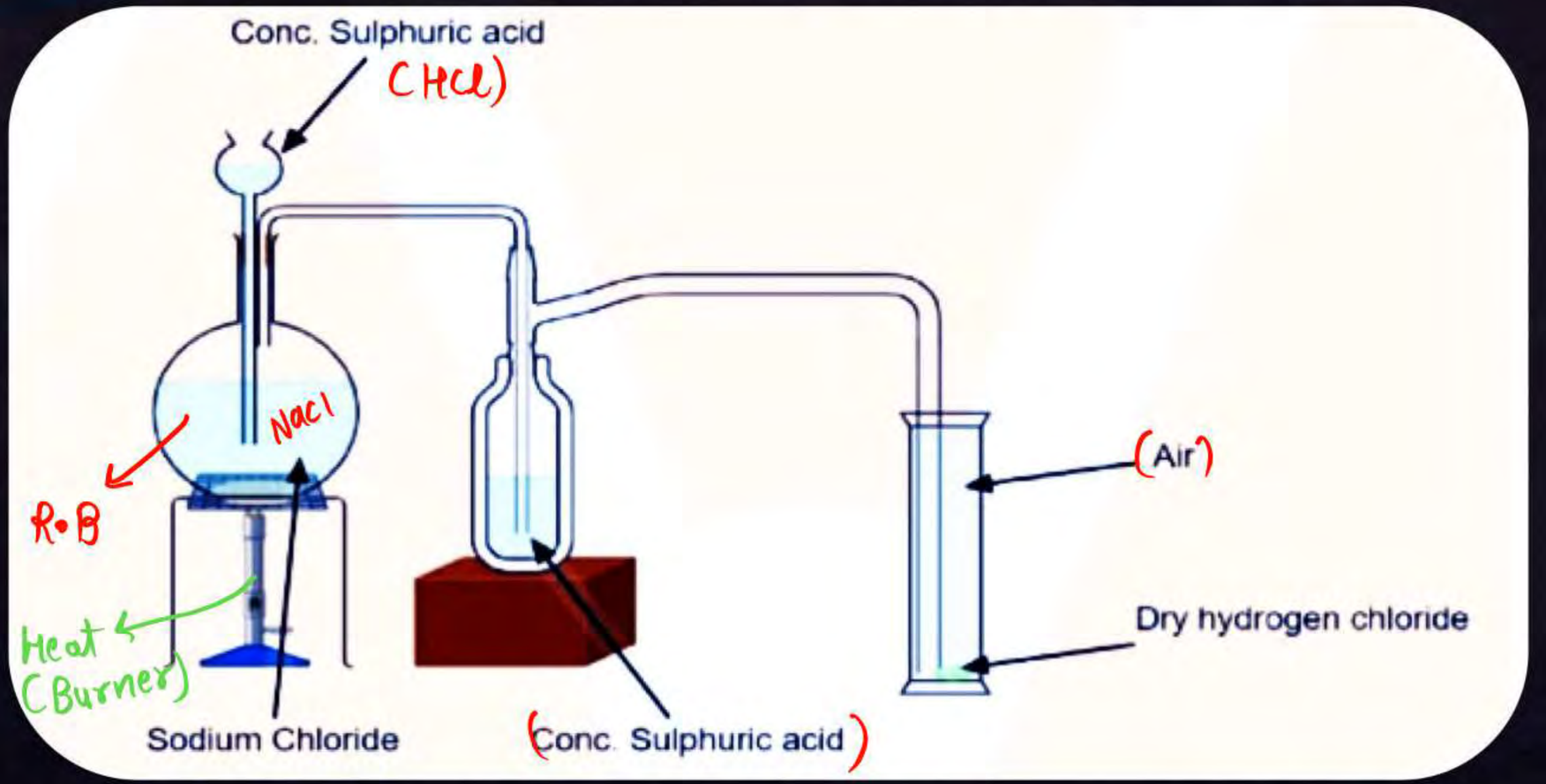
Sodium chloride and concentrated sulphuric acid.

Procedure:

Set up the apparatus as shown

$$NaCl + H_2SO_4 \xrightarrow{200^\circ C} NaHSO_4 + HCl$$

Laboratory preparation of hydrogen chloride gas



Laboratory preparation of hydrogen chloride gas

1. Place some common salt in a flask and pour concentrated sulphuric acid through the this the funnel

2. On heating the mixture gently there is effervescence and HCl gas is evolved in a controlled manner

3. The gas is dried by passing through a washer bottle containing I concentrated sulphuric acid.

Conc. H_2SO_4 → V. good drying and dehydrating agent

Purification of HCl gas:

1. It is dried by passing through conc. Sulphuric acid.

Imp. PYQ

2. The other drying agents like phosphorus pentoxide (P_2O_5) and quick lime (CaO) cannot be used, since they react with hydrogen chloride.

$$2P_2O_5 + 3HCl \rightarrow POCl_3 + 3HPO_3$$

(Phosphorus oxychloride)

$$CaO + 2HCl \rightarrow CaCl_2 + H_2O$$

Collection of HCl gas

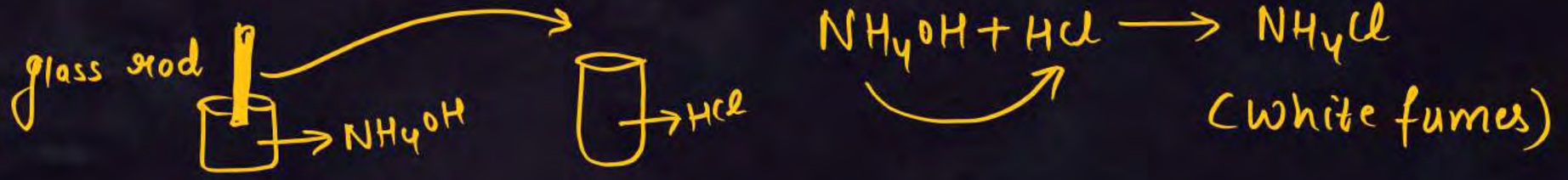


→ By upward displacement of air

density of HCl > density of air

Identification

1. In order to know whether the gas jar is full. Bring a rod dipped in ammonium hydroxide near its mouth.



2. Dense white fumes of ammonium chloride will be produced proving thereby that the jar is full of hydrogen chloride gas.



Precautions

a) The lower end of the thistle funnel must be dipped in conc. sulphuric acid.

b) Delivery tube should be dipped in drying agent i.e., (conc. H_2SO_4.)

c) Temperature is maintained at nearly(200°C)



Physical properties of hydrogen chloride gas

1.	Colour	Colourless gas ✓
2.	Smell	Pungent choking smell ✓
3.	Taste	Sour (acidic) in taste ✓
4.	Physiological nature	It is corrosive in nature It irritates nose, throat and lungs
5.	Density	It is about one and a quarter times heavier than air (V.D. of HCl is 18.25 and that of air 14.4). $\rho_{HCl} > \rho_{air}$

Physical properties of hydrogen chloride gas

6. Boiling point -83°C ✓

7. Melting point -113°C ✓



Toluene, Acetone (organic solvent)

8. Liquefaction and solidification

↓ High pressure and low temp

When subjected to a high pressure (40 atm.) at low temperature (10°C), it is liquefied to a colourless liquid.

9. Solubility

HCl being polar covalent compound is soluble in organic! non-polar solvents also such as acetone and toluene.

Physical properties of hydrogen chloride gas

Experiment to demonstrate density:

To show that HCl gas is heavier than air.

1. It can be proved by pouring the gas in a jar with a burning candle.

2. The candle gets, extinguished) because HCl gas being heavier occupies the lower portion of the jar and forces the air out of it.

3. This experiment also combustible nor a supporter of that HCl gas is neither f combustion.

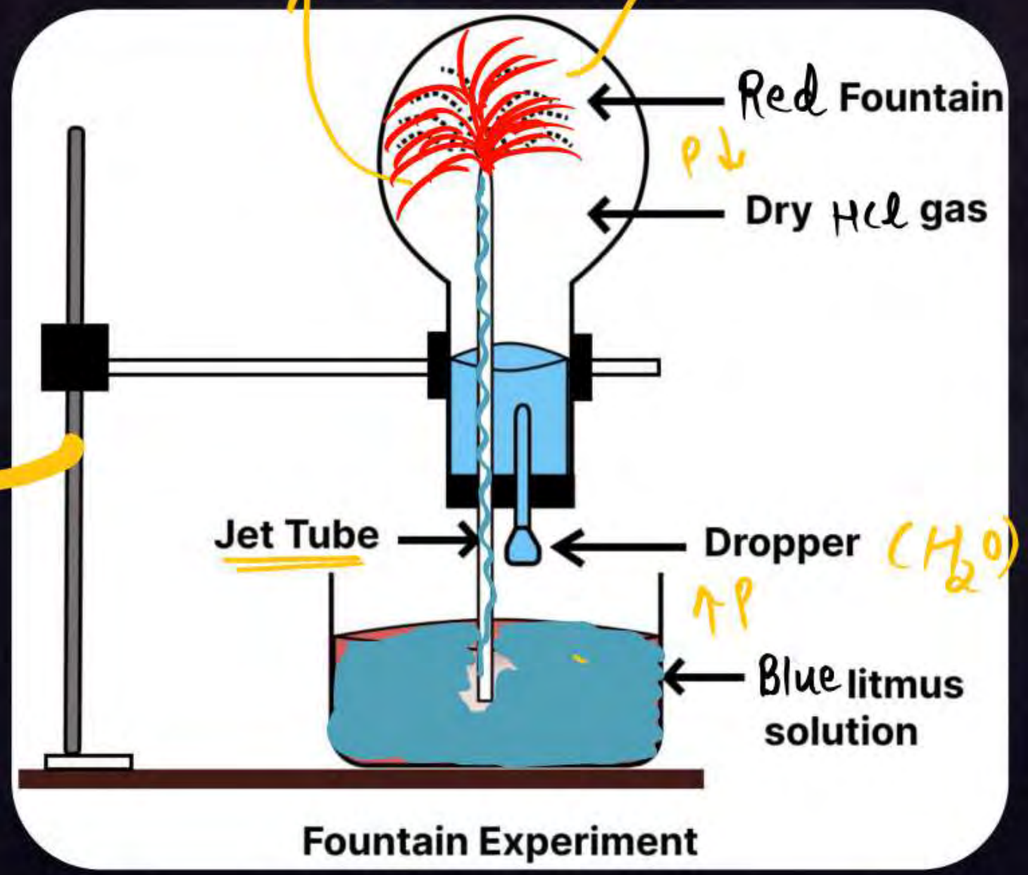




Fountain Experiment

1. Take a (dry round bottomed) flask filled with dry HCl gas
2. On the mouth of the flask, fix a rubber stopper with two holes, with
3. One hole pass a long jet tube and through other hole pass a dropper with few drops of water.
4. Put the jet tube in the beaker containing blue litmus solution
5. Press the dropper, it is seen that the blue litmus solution enters the jet tube with a great force forming a red fountain.

Fountain Experiment



Dry HCl(g)

R.B

Clamp Stand

P↓

↑P

(H_2O)

R.B = HCl(g) High pressure
+
via dropper, water

$HCl(g) \rightarrow HCl(liq)$
Acid

↳ due to the formation of HCl(acid)
Pressure dec.↓

HCl → Acidic

Acid; Convert Blue Litmus soln

↓ to

Red Litmus solution

Doubts

R.B → low pressure

Outside R.B → High Pressure

Push Blue litmus in Jet tube

Dropper :→ Water

(R.B)

HCl (g) ⇝

Highly soluble in H_2O

(HCl) liq

Pressure dec. ↓ in (R.B)

←

Which form diff in the Pressure

Chemical properties of hydrogen chloride gas

1. Combustibility: The gas is neither combustible nor a supporter of combustion. It does not burn, rather it extinguishes a burning splint.

2. Thermal dissociation: On heating above 500°C, it dissociates into hydrogen and chlorine.

$$2HCl \xrightarrow{<500^{o}C} H_2 + Cl_2$$

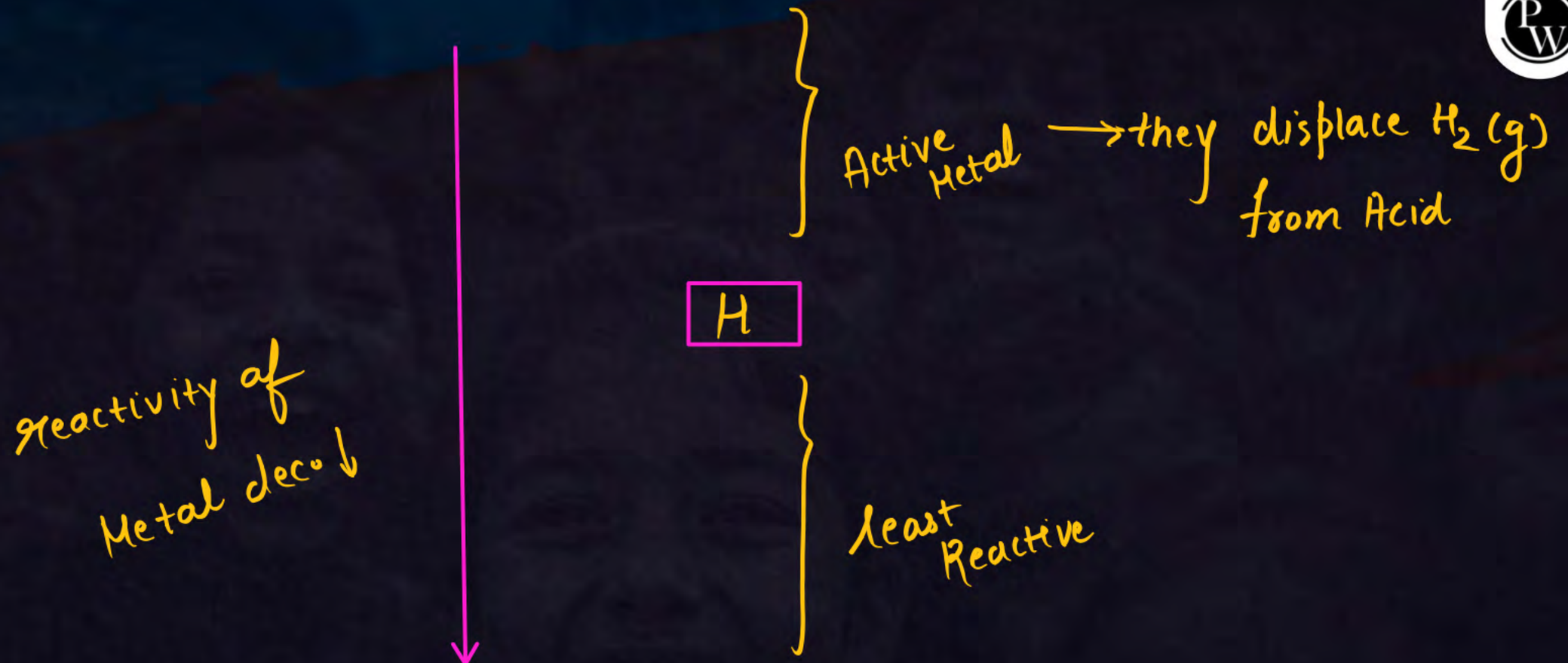
Chemical properties of hydrogen chloride gas

3. With metals: Metals that ~~com~~ is present before hydrogen in the electrochemical series form their corresponding chlorides when heated with HCl and liberate hydrogen gas ($H_2\uparrow$).

Metal + Hydrogen chloride (g) → Metal chloride + Hydrogen

$2Na + 2HCl \rightarrow 2NaCl + H_2\uparrow$

$Ca + 2HCl \rightarrow CaCl_2 + H_2\uparrow$


Active Metal → they displace $H_2(g)$ from Acid
H
Least Reactive
Reactivity of Metal dec. ↓

Chemical properties of hydrogen chloride gas

1. Reaction with ammonia: It combines with ammonia to form dense white fumes of ammonium chloride.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

i.e., two gases combine to form solid

Imp

test for HCl(g)



Chemical Properties

2. Action on metals (Action of HCl on Metal)

Metal + Hydrogen Chloride → Metal chloride + Hydrogen

$Ca + 2HCl \rightarrow CaCl_2 + H_2\uparrow$

$Mg + 2HCl \rightarrow MgCl_2 + H_2\uparrow$



Chemical Properties

Note:- Metal oxide are Basic in Nature

3. Action on oxides and hydroxide:

Acid + Base → Salt + H_2O

(Base)

Metal oxide/ hydroxide + Hydrochloric acid (Acid) → Metal chloride + H_2O

$MgO + 2HCl \rightarrow MgCl_2 + H_2O$

$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$



Chemical Properties

4. Which salts of weaker acids: (Reaction of HCl with Metal Carbonate)

Metal/ hydrogen + Hydrochloric acid → Metal chloride salt + carbon dioxide + H_2O

carbonates

Na_2CO_3 + HCl → NaCl + H_2O + $CO_2\uparrow$

$CaCO_3$ + HCl → $CaCl_2$ + H_2O + $CO_2\uparrow$

(Acid)

(slightly Basic)

Concept:
Jab-Jab Reactant mai CO_3^{2-} / HCO_3^- } → Product main $CO_2(g)\uparrow$

Concept:- Metal Carbonate / Metal hydrogen carbonate are slightly Basic



Chemical Properties

5. Precipitation Reaction

Normally react with nitrates. However lead nitrate react with hydrochloric acid to given white precipitate of lead chloride and mercury (I) chloride.

$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2 \downarrow + 2HNO_3$

$Hg_2(NO_3)_2 + 2HCl \rightarrow Hg_2Cl_2 \downarrow + 2HNO_3$

Precipitate of lead chloride dissolves on heating

white

$PbCl_2$ (ppt)

Hg_2Cl_2 (Solid Insoluble Mass)

Uses of Hydrochloric acid

I. General uses:

In the laboratory as a reagent and for preparation of aqua regia.

Aqua Regia is very-very Corrosive reagent which can dissolve gold, Pt

also known as: 'Royal water'



Uses of Hydrochloric acid

II) Industrial uses:

1. In the manufacture of

   a) Chlorine and chlorides e.g., ammonium chloride used in dry cells.

   b) Dyes, drugs, paints and photographic chemicals (silver chloride).

   c) Glucose from starch.



Uses of Hydrochloric acid

II) Industrial uses:

2. In industry to pickle* steel (clean metal surface by using acid), as HCl dissolves the oxides. Steel before being plated with tin or chromium has to be purified by pickling in HCl and some inhibitors.



Uses of Hydrochloric acid

II) Industrial uses:

3. For purifying bone black, because HCl dissolves the calcium phosphate present in bones.



Uses of Hydrochloric acid

II) Industrial uses:

4. To remove rust from iron sheets.



Uses of Hydrochloric acid

II) Industrial uses:

5. For cleaning metal surfaces before painting, electroplating, galvanizing, soldering, etc.



Uses of Hydrochloric acid

II) Industrial uses:

6. In the extraction of glue from bones.



Uses of Hydrochloric acid

II) Industrial uses:

7. In tanning and calico printing industry.



Uses of Hydrochloric acid

II) Medicine:

Dilute HCl is prescribed to patients with decreased activity of their gastric juices.

Question

Nature of MgO, CuO is:



A Amphoteric → ZnO, Al_2O_3

B Acidic

C Basic ✓

D Neutral

Metal oxide
↳ Basic

Tests for hydrogen chloride gas and Hydrochloric acid

1. HCl gas possesses a characteristic irritating smell. Imp

2. HCl gas gives thick white fumes of ammonium chloride, when a glass rod dipped in ammonia solution is held near the vapours of the acid.

$NH_3 + HCl \rightarrow NH_4Cl$ (White fumes)

Tests for hydrogen chloride gas and Hydrochloric acid

3. With silver nitrate solution, both the gas and the acid give a white precipitate of silver chloride.

$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$

(white)

Tests for hydrogen chloride gas and Hydrochloric acid

4. The precipitate is insoluble in nitric acid but soluble in ammonium hydroxide. A greenish-yellow gas, i.e. chlorine, is liberated when concentrated hydro-chloric acid is heated with an oxidising agent like manganese dioxide.

$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2\uparrow$

The gas liberated turns moist starch iodide paper blue black.

$MnO_2 + HCl \longrightarrow MnCl_2 + H_2O + Cl_2(g)$

$Cl_2 \rightarrow$ greenish yellow color

Q Type of Reaction } How

Que Reducing Agent ? }

Question

Name the glassware labelled X in the following figure.

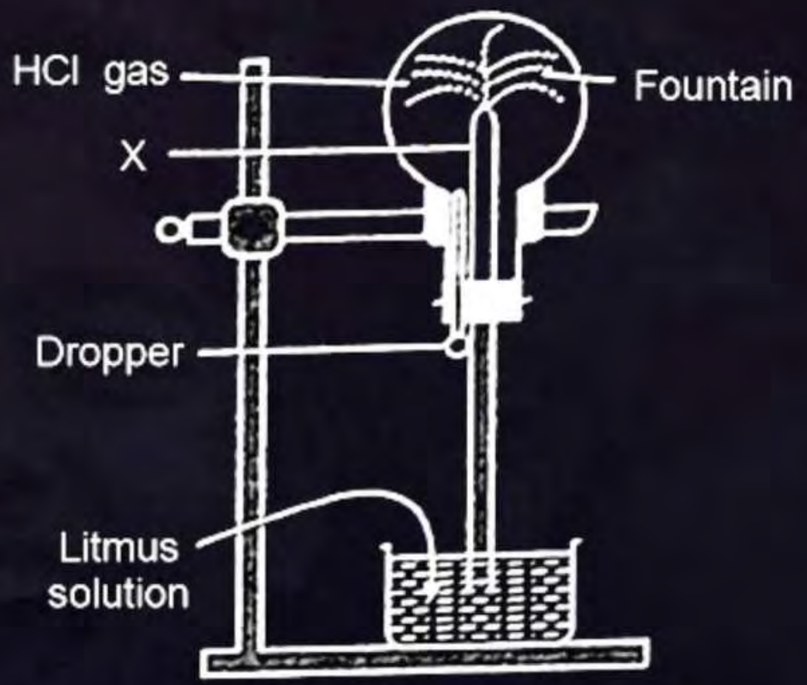


A Trough

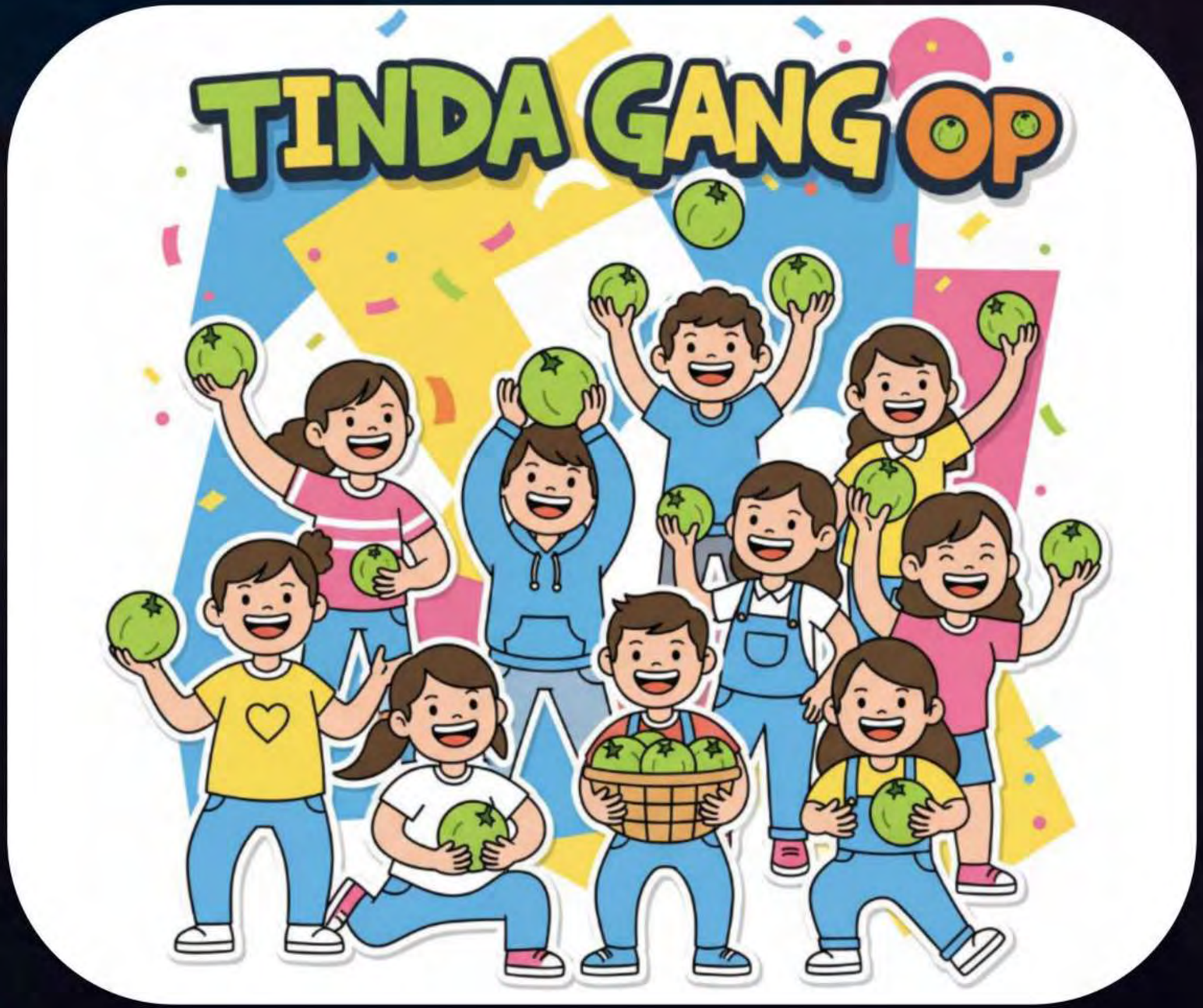
B Jet tube

C Glass rod

D Plastic pipe




BHARTI MAAM
JOIN MY OFFICIAL TELEGRAM CHANNEL
Physics Wallah
@CHEMISTRYBYBHARTIMAAM


TINDA GANG OP


Thank You